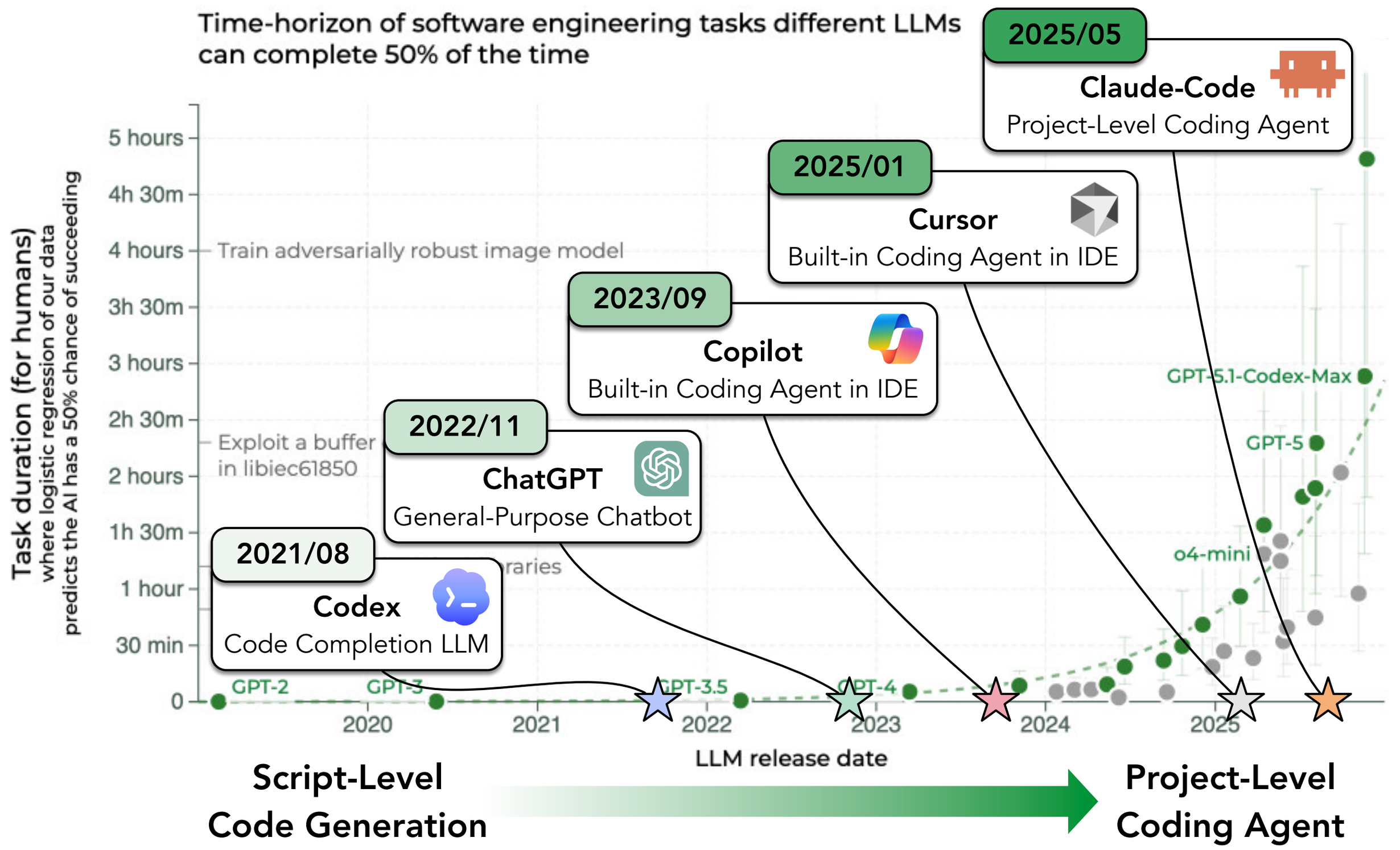
Time-horizon of software engineering tasks different LLMs can complete 50% of the time
Task duration (for humans)
where logistic regression of our data predicts the AI has a 50% chance of succeeding
5 hours
4h 30m
4 hours
3h 30m
3 hours
2h 30m
2 hours
1h 30m
1 hour
30 min
0
Train adversarially robust image model
Exploit a buffer in libiec61850
2021/08
Codex
Code Completion LLM
2022/11
ChatGPT
General-Purpose Chatbot
2023/09
Copilot
Built-in Coding Agent in IDE
2025/01
Cursor
Built-in Coding Agent in IDE
2025/05
Claude-Code
Project-Level Coding Agent
GPT-2
GPT-3
GPT-3.5
GPT-4
o4-mini
GPT-5
GPT-5.1-Codex-Max
2020
2021
2022
2023
2024
2025
LLM release date
Script-Level Code Generation
Project-Level Coding Agent